

Sydney Girls High School



MATHEMATICS EXTENSION 1

Year 11

Task 2 (Half-yearly)

April 2015

Time Allowed: 60 minutes + 5 minutes reading time

Topics: Basic Arithmetic & Algebra; Inequalities; Real Functions; Trigonometric ratios

Total: 64 marks

Instructions:

- There are FOUR (4) questions which are of equal value.
- Attempt all questions.
- Show all necessary working. Marks will be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write your student number at the top of each question and clearly number each question.

Student's Name:_____ **Teacher's Name:**_____

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Question 1: (16 Marks)

a) If $f(x) = 3x^2 - 5x + 4$ and $g(x) = 2x + 10$ find $f(1) - g(-1)$. 2

b) Find the exact value of $\sin \theta$ if $\tan \theta = \frac{3}{7}$ and $\cos \theta < 0$. 3

c) Find the integers a and b such that $\frac{2}{2 - \sqrt{3}} = a + \sqrt{b}$. 2

d) Express $\frac{x+1}{x^2-x} - \frac{x-1}{x^2+x}$ as a fraction in its lowest terms. 3

e) Solve $\frac{2}{x+3} \leq 5$ 3

f) Find the range of the function $y = 8x - x^2$ 1

g) Expand and simplify $(2x - y)^4$ 2

Question 2: (16 Marks)

a) Find the value of m if $\sec 27^\circ = \operatorname{cosec}(m + 20)^\circ$ 1

b) Determine if the function $f(x) = x^3 - x^2$ is odd, even or neither. 2
Show working to justify your answer.

c) Sketch the following showing all relevant features.

i) $2x - y = 1$ 1

ii) $y = 3 - x^2$ 1

iii) $y = (2x - 1)^3$ 1

iv) $y = -2^x$ 1

v) $y = |x^2 - 1|$ 1

vi) $y = \frac{2}{x^2} + 1$ 2

d) On the same diagram shade the following region for which $y \leq x^2$, $y \geq 0$ and 3
 $y \leq \sqrt{9 - x^2}$ are simultaneously true.

e) Find the value of x and y if $2^{2x+y} = 128$ and $\frac{4^x}{16} = 8^{x+y}$ 3

Question 3: (16 Marks)

a) A function is defined by the rule $f(x) = \begin{cases} 1 & \text{if } x \leq -2 \\ -1 & \text{if } -2 < x < 0 \\ x & \text{if } x \geq 0 \end{cases}$

i) Sketch the curve 2

ii) Find $f(a^2)$ 1

b) i) Show that $\frac{x+4}{x+3} = 1 + \frac{1}{x+3}$. 1

ii) Find the domain and range of the function 2

$$y = \frac{x+4}{x+3}$$

iii) Sketch the graph of $y = \frac{x+4}{x+3}$ 2

c) Solve the following equations where $0 \leq \theta \leq 360^\circ$

i) $\cos \theta = 0.7$ 2

ii) $\sin 2\theta = -\frac{\sqrt{3}}{2}$ 2

d) State the domain of the function $h(x) = \sqrt{2+x} - \sqrt{3-x}$ 2

e) On the same graph sketch the following for $0 \leq x \leq 360^\circ$ 2

i) $y = \sin x$

ii) $y = \operatorname{cosec} x$

Question 4: (16 Marks)

- a) From a ship travelling north, a tower on shore is observed to be on a bearing of 37° . Later when the ship is 2 nautical miles from the tower, the bearing of the tower from the ship is 127° .

- i) Draw a diagram marking on it all the information given. 1
- ii) Find, correct to 3 significant figures, the distance of the tower from the initial point of observation. 2

b) Prove $\frac{\tan x}{\sec x - 1} = \frac{\sec x + 1}{\tan x}$. 2

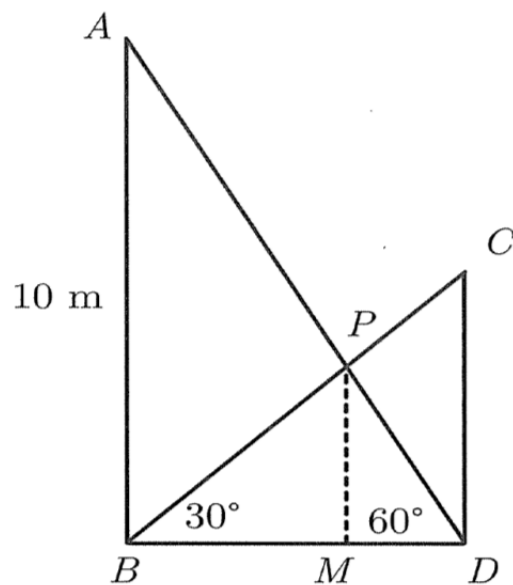
- c) i) Sketch the curves $y = |x - 3|$ and $y = 5 - |x|$ on the same diagram. 2

- ii) Hence or otherwise solve, $|x - 3| + |x| < 5$ 1

d) Solve $6 \tan \theta - 5 \operatorname{cosec} \theta = 0$ for $0 \leq \theta \leq 360^\circ$. 4

Question 4(continued): (16 Marks)

- e) AB is a vertical pole height 10m. CD is a shorter vertical pole of unknown height. The angle of elevation from the base of AB to the top of CD is given as 30° and the angle of elevation from the base of CD to the top AB is given as 60° . Another vertical pole PM is placed at the point of intersection of the angles of elevation.



- | | | |
|------|---|---|
| i) | Copy the diagram onto your exam paper | |
| ii) | Find the exact distance the poles AB and CD are apart. | 1 |
| iii) | Find the exact height of CD | 1 |
| iv) | Find the exact height of PM | 2 |

THE END

2015 YR EXT TASK 1

Q1

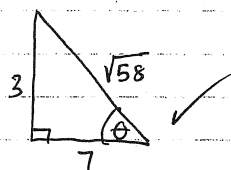
a) $f(x) = 3x^2 - 5x + 4$

$g(x) = 2x + 10$

Find $f(1) - g(-1)$
 $= 3(1)^2 - 5(1) + 4 - (2(-1) + 10)$
 $= -6$ ✓

b) $\tan \theta > 0$
 $\cos \theta < 0$ ∴ 3rd quadrant $\pi < \theta < \frac{3\pi}{2}$ ✓

$\tan \theta = \frac{3}{7}$



∴ $\sin \theta = -\frac{3}{\sqrt{58}}$ ✓

c) $\frac{2}{2-\sqrt{3}} = a + \sqrt{b}$

$\frac{2(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})} = \frac{4+2\sqrt{3}}{1} = a + \sqrt{b}$ ✓

∴ $a = 4$
 $4 \times 3 = b$ ∴ $b = 12$ ✓

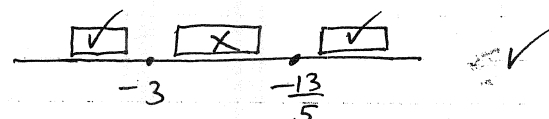
d) $\frac{x+1}{x(x-1)} - \frac{x-1}{x(x+1)}$

$\frac{(x+1)^2 - (x-1)^2}{x(x-1)(x+1)} = \frac{x^2+2x+1 - x^2+2x-1}{x(x-1)(x+1)}$ ✓
 $= \frac{4}{(x-1)(x+1)}$ ✓

e) $\frac{2}{x+3} \leq 5$

Critical points: $x = -3$

$\frac{2}{x+3} = 5 \quad \therefore 2 = 5x + 15$
 $x = -\frac{13}{5}$



Try $x = -4$: $-2 \leq 5$ True

Try $x = -2.8$: $10 \leq 5$ False ✓

Try $x = -1$: $1 \leq 5$ False

Final solutions

$x < -3$ OR $x > -\frac{13}{5}$ ✓

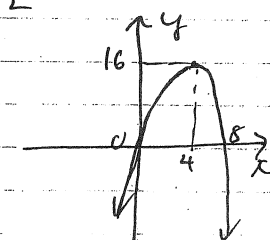
f) $y = 8x - x^2 = -x(x-8)$

vertex: $x = \frac{-b}{2a} = \frac{-8}{2(-1)} = 4$

$y = 8(4) - (4)^2$

$y = 16$

Range: $y \leq 16$ ✓



g) $(2x-y)^4 = (2x)^4 - 4(2x)^3(y) + 6(2x)^2y^2 - 4(2x)y^3 + y^4$ ✓
 $= 16x^4 - 32x^3y + 24x^2y^2 - 8xy^3 + y^4$ ✓

~~XXXXXX~~

Question 2

(a) $90^\circ - 27^\circ = 63^\circ$ 1

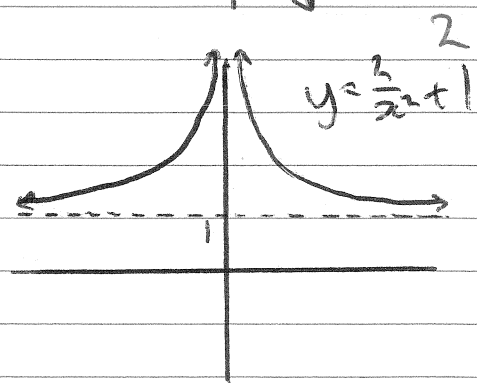
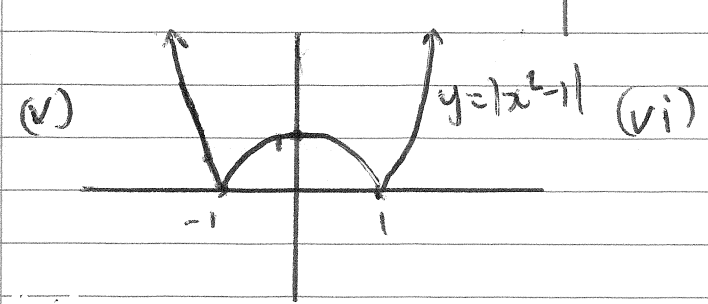
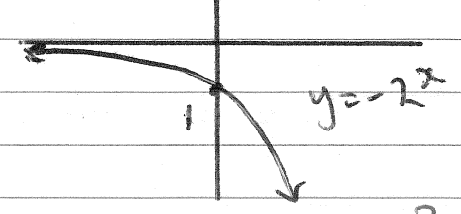
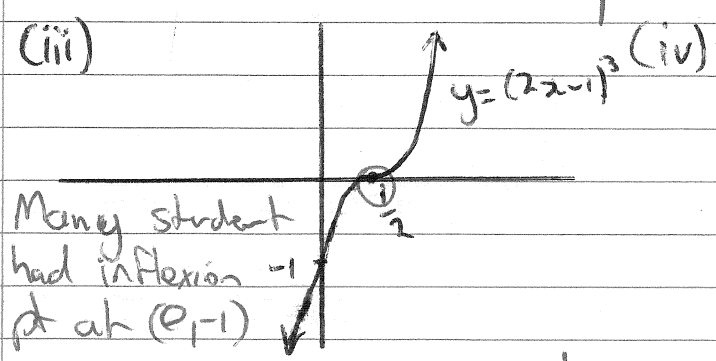
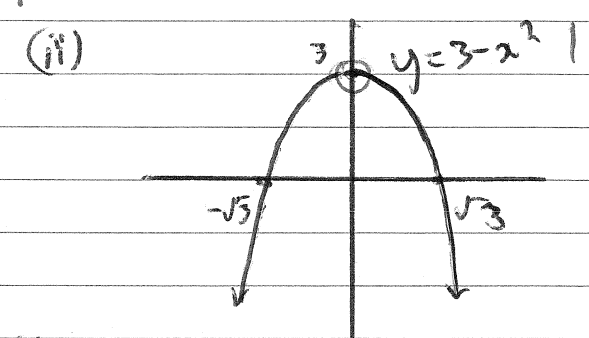
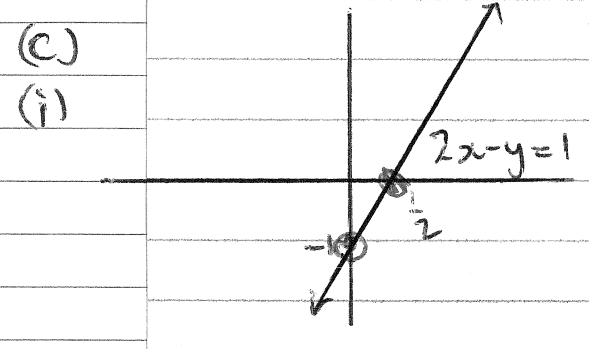
$m + 20^\circ = 63^\circ$

$m = 43^\circ$

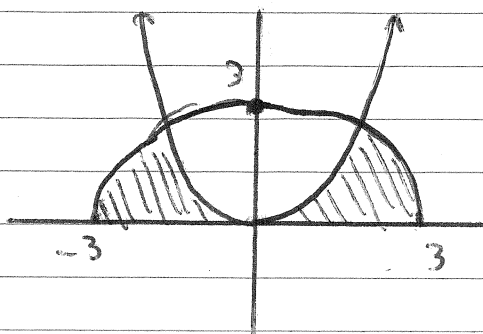
(b) $f(-x) = (-x)^3 - (-x)^2$

$= -x^3 - x^2$
 $\neq f(x)$ or $-f(x)$.

NEITHER



(d)



3

Most Student shaded
above the parabola
 $y = x^2$.

$$(e) \quad 2^{2x+y} = 128$$

$$2^{2x+y} = 2^7$$

$$2x + y = 7 \quad (A)$$

Sub (B) into (A)

$$-6y - 8 + y = 7$$

$$-5y - 8 = 7$$

$$-5y = 15$$

$$y = -3$$

Sub into (B)

$$x = 9 - 4$$

$$x = 5$$

$$x = 5, y = -3$$

$$\frac{4^x}{16} = 8^{x+y}$$

$$\frac{2^{2x}}{2^4} = 2^{3x+3y}$$

$$2^{2x-4} = 2^{3x+3y}$$

$$2x - 4 = 3x + 3y$$

$$-x = 3y + 4 \quad (B)$$

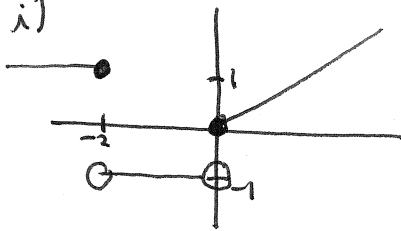
$$x = -3y - 4$$

3

Many Student simplified

$$\frac{2^{2x}}{2^4} \text{ as } 2^{\frac{2x}{4}}!$$

3a) i)



Open and closed circles must be clearly shown

ii) $f(a^2) = a^2$

$$\begin{aligned} \text{iv) i) } \frac{x+4}{x+3} &= \frac{x+3+1}{x+3} \\ &= \frac{x+3}{x+3} + \frac{1}{x+3} \\ &= 1 + \frac{1}{x+3} \quad \text{G.E.D.} \end{aligned}$$

ii) domain: $x \neq -3$

range: $y \neq 1$

v) i) $\theta = 46^\circ$ or $360^\circ - 46^\circ$
 $= 46^\circ$ or 314°

ii) $2\theta = 180^\circ + 60^\circ, 360^\circ - 60^\circ,$
 $540^\circ + 60^\circ, 720^\circ - 60^\circ$
 $= 240^\circ, 300^\circ, 600^\circ, 660^\circ$
 $\theta = 120^\circ, 150^\circ, 300^\circ, 330^\circ$

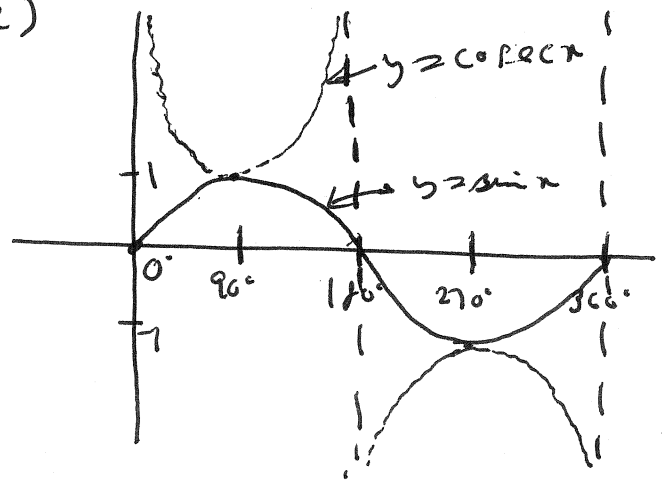
$[180^\circ \leq \theta \leq 360^\circ \text{ then } 0^\circ \leq 2\theta \leq 720^\circ]$

d) $2+x \geq 0$ and $3-x \geq 0$
 $x \geq -2$ $-x \geq -3$
 $x \leq 3$

$\therefore -2 \leq x \leq 3$

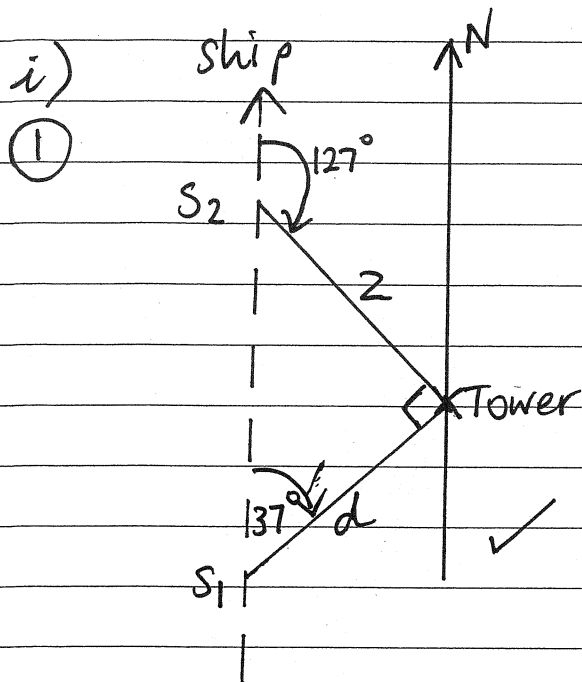
Some forgot to combine the solutions.

e)



Any asymptotes must be drawn for $y = \csc x$.

Question 4 (16 Marks)



ii) (2)

$$\tan 37^\circ = \frac{2}{d} \quad \checkmark$$
$$\therefore d = \frac{2}{\tan 37^\circ}$$
$$d \doteq 2.65 \text{ nm} \quad \checkmark$$

(3 sig. figs)

ii)

(2)

$$\text{LHS} = \frac{\tan x}{\sec x - 1}$$

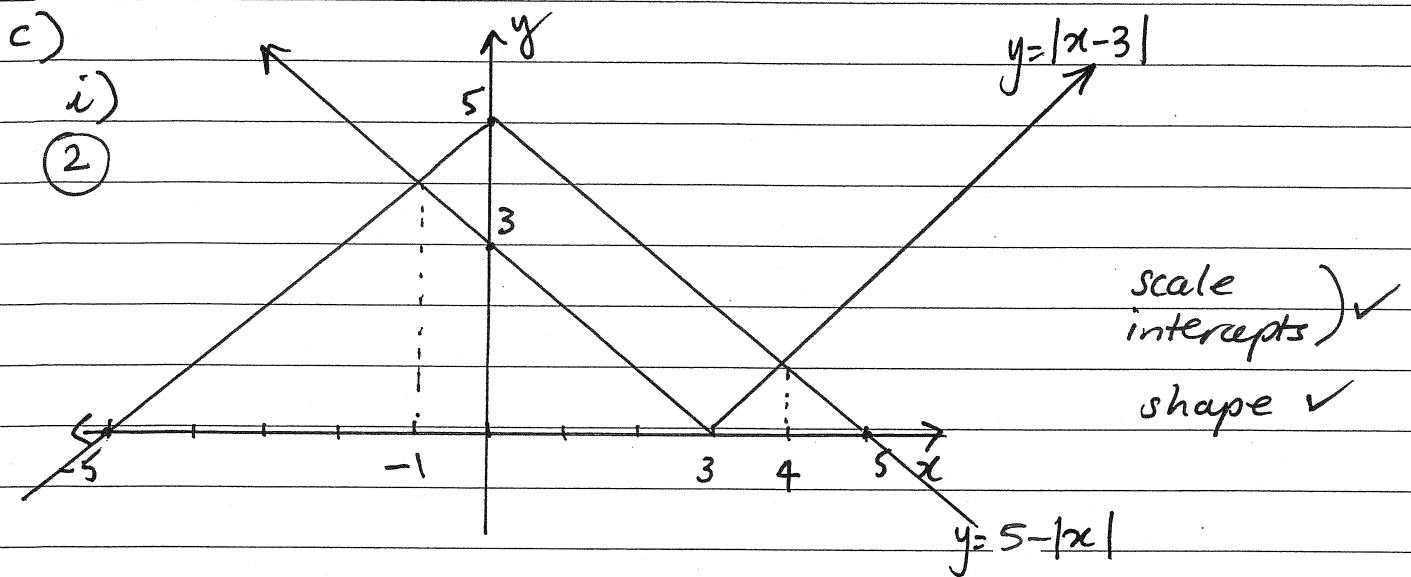
$$= \frac{\tan x}{(\sec x - 1)} \times \frac{(\sec x + 1)}{(\sec x + 1)}$$

$$= \frac{\tan x (\sec x + 1)}{\sec^2 x - 1} \quad \checkmark$$

$$= \frac{\tan x (\sec x + 1)}{\tan^2 x} \quad \checkmark$$

$$= \frac{(\sec x + 1)}{\tan x}$$

$$= \text{RHS}$$



ii) $|x-3| + |x| < 5$

(1) $|x-3| < 5-|x|$

$\therefore -1 < x < 4$ (from graph). ✓

d) $6 \tan \theta - 5 \operatorname{cosec} \theta = 0$, $0 \leq \theta \leq 360^\circ$.

(4) $6 \frac{\sin \theta}{\cos \theta} - 5 \cdot \frac{1}{\sin \theta} = 0$ ✓ ($\sin \theta \neq 0, \cos \theta \neq 0$)

$$6 \sin^2 \theta - 5 \cos \theta = 0$$

$$6(1 - \cos^2 \theta) - 5 \cos \theta = 0$$
 ✓

$$\therefore 6 \cos^2 \theta + 5 \cos \theta - 6 = 0$$

$$6 \cos^2 \theta + 9 \cos \theta - 4 \cos \theta - 6 = 0$$

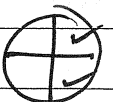
$$3 \cos \theta (2 \cos \theta + 3) - 2(2 \cos \theta + 3) = 0$$

$$(2 \cos \theta + 3)(3 \cos \theta - 2) = 0$$

$$\therefore 2 \cos \theta + 3 = 0 \quad \text{or} \quad 3 \cos \theta - 2 = 0$$

$$\cos \theta = -\frac{3}{2}$$

$$\cos \theta = \frac{2}{3}$$

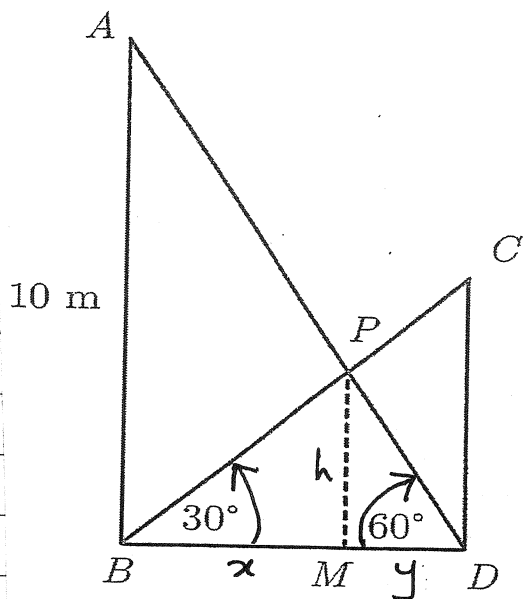


$$\theta = \text{No solutions}$$

$$\theta = 48^\circ 11', 311^\circ 49'$$



e) i)



ii) Find BD:

(1)

$$\text{In } \triangle ABD: \tan 60^\circ = \frac{AB}{BD}$$

$$\therefore BD = \frac{AB}{\tan 60^\circ}$$

$$\therefore BD = \frac{10}{\sqrt{3}} = \frac{10\sqrt{3}}{3} \text{ m.}$$

iii) Find CD:

(1)

$$\text{In } \triangle CDB: \tan 30^\circ = \frac{CD}{BD}$$

$$\frac{1}{\sqrt{3}} = \frac{CD \times \sqrt{3}}{10}$$

$$\therefore CD = \frac{1}{\sqrt{3}} \times \frac{10}{\sqrt{3}} = \frac{10}{3} \text{ m}$$

iv) Find PM. Let $BM = x$, $MD = y$ and $PM = h$.

(2)

$$\text{In } \triangle PMB: \tan 30^\circ = \frac{h}{x}$$

$$\text{In } \triangle PMD: \tan 60^\circ = \frac{h}{y}$$

$$\therefore x = \frac{h}{\tan 30^\circ}$$

$$\therefore y = \frac{h}{\tan 60^\circ}$$

$$\text{Now: } x + y = \frac{h}{\tan 30^\circ} + \frac{h}{\tan 60^\circ}$$

$$BD = \sqrt{3}h + \frac{h}{\sqrt{3}}$$

$$\text{from ii) } \frac{10}{\sqrt{3}} = \left(\frac{3+1}{\sqrt{3}} \right) h$$

$$\therefore h = \frac{10}{\sqrt{3}} \times \frac{\sqrt{3}}{4} = \frac{10}{4}$$

$$\therefore h = 2.5 \text{ metres.}$$